

ME 103

Final Project

Thermal Response of
TMPS Lattice Structures

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VOTING
IN ELECTIONS



WALKING
YOUR DOG



SOME
THIRD OPTION

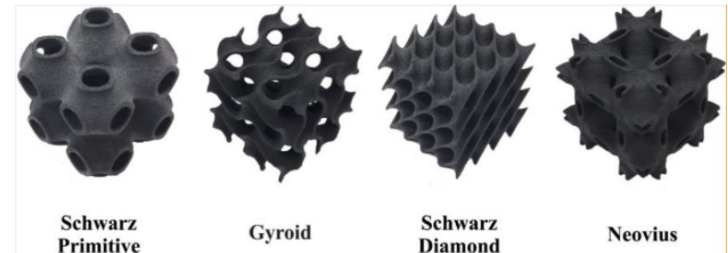
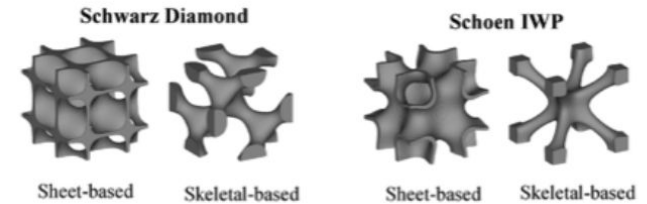
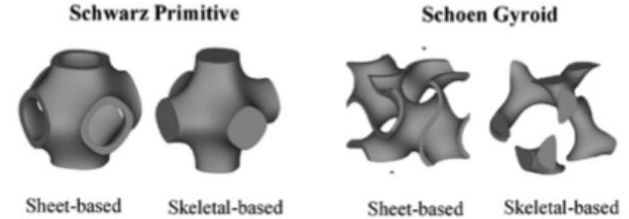


DOING ME 103
HEAT TRANSFER
PROJECT



Motivation & Objectives

- Manufacturing Innovation: Advancements in metal AM enable the fabrication of complex TPMS geometries.
- Thermal Performance: TPMS architectures offer high surface-area to-volume ratios and induce vortical flow patterns that disrupt boundary layers.
 - SA:V of the block: 0.168, sleeves 1.178, GRY 2.108, FRD 3.259
- Weight Optimization: The continuous nature of TPMS lattices minimizes pressure drop while providing structural integrity.



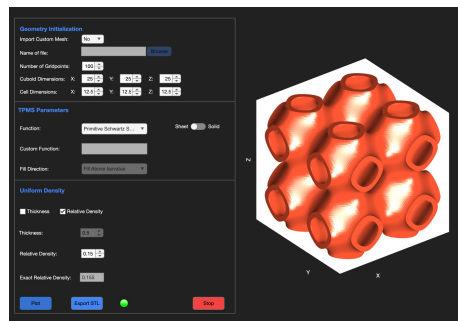
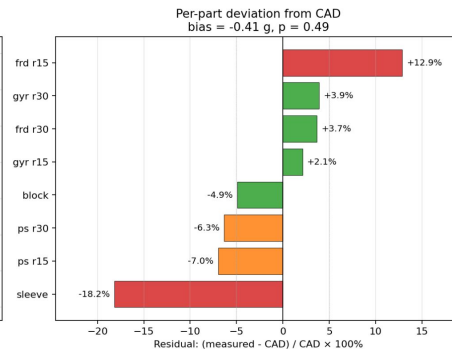
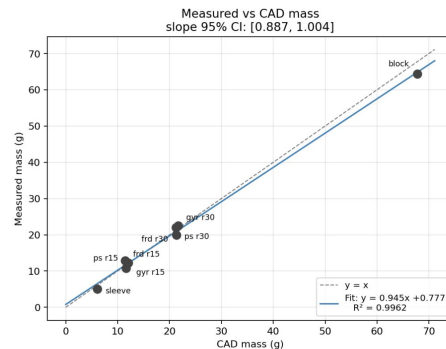
Hypothesis

- **Null hypothesis (H_0):** Lattice topology has no effect on through-thickness heat transfer at fixed relative density. All topologies (Gyroid, FRD, PS) produce equivalent back-face temperatures and ΔT responses within measurement uncertainty.
- **Alternative Hypothesis (H_1):** Lattice topology produces measurably different heat-transfer responses at fixed density.

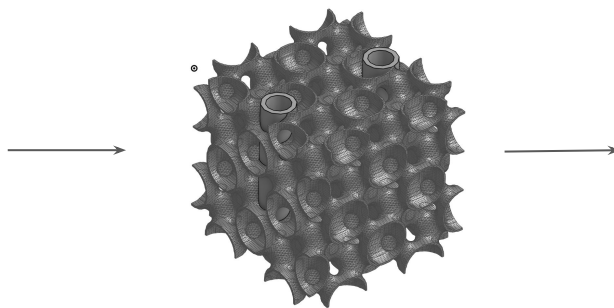


Sample Preparation

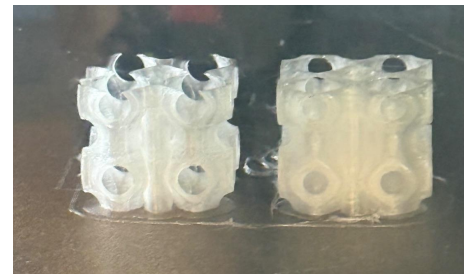
- Open sourced lattice generator → stl (LatGen)
- Modify in CAD for thermocouple integration
- FDM print int transparent PLA
- Record mass



latgen



CAD



printing

Uncertainty Summary

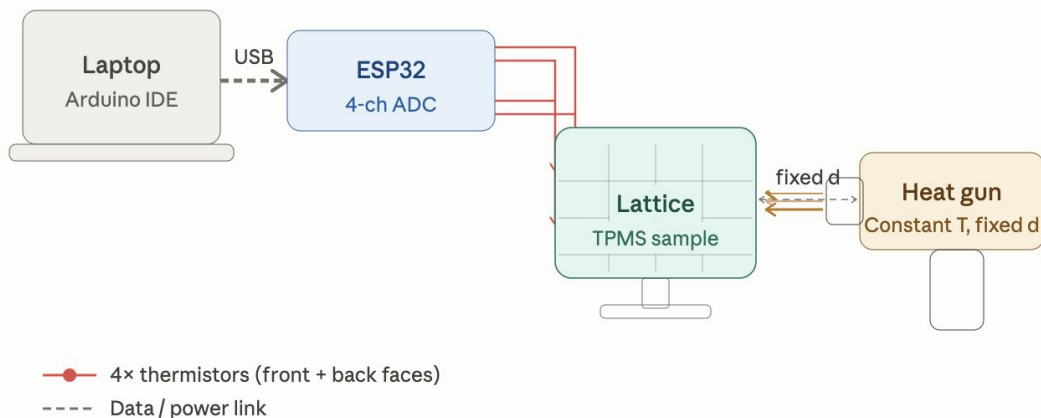
Combined measurement uncertainty: ± 0.31 °C and ± 1.12 °C respectively.

All ΔT comparisons between geometries exceed these uncertainty bounds. *Conduction temp differences are slightly smaller

DOE 1 | Conjugate Heat Transfer Study

Conjugate heat transfer — experimental setup

ESP32 · 4× thermistors · Arduino IDE · heat gun at fixed distance



DOE 1 | Conjugate Heat Transfer Study

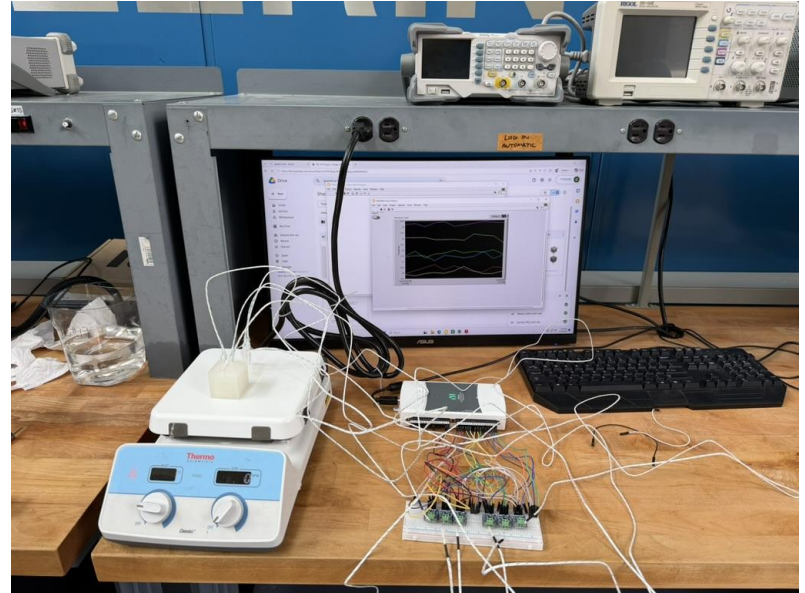
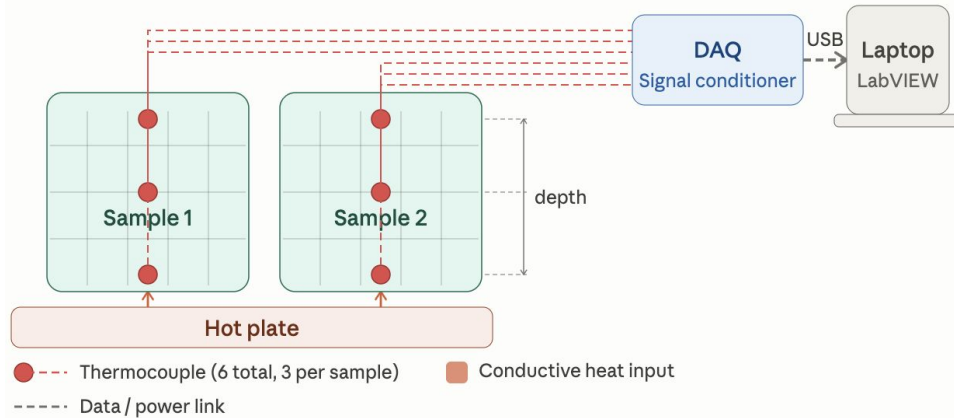
VARIABLE TABLE

Independent	Dependent	Control
Geometry (Gyroid, FRD, Primitive Schwartz, Solid Block, Sleeve)	Front-face temperature T_f (°C)	Heat gun nozzle distance
Relative density (RD = 0.15, 0.30)	Back-face temperature T_b (°C)	Ambient room temp.
	$\Delta T = T_f - T_b$ (°C)	Sample material & dimensions

DOE 2 | Conduction Study

Conduction study — experimental setup

6 thermocouples · 3 per sample · DAQ · LabVIEW



DOE 2 | Conduction Study

VARIABLE TABLE

Independent	Dependent	Control
Lattice geometry (Gyroid, FRD)	Temperature vs. time at each depth $T(t)$ ($^{\circ}\text{C}$)	Hot plate set temperature (confirmed by TC plateau)
Thermocouple depth position (1/4, 1/2, 3/4)	Steady-state temperature T_{ss} ($^{\circ}\text{C}$)	Sample material
	Depth-to-depth ΔT ($^{\circ}\text{C}$)	TC type (Type K)

Calibration

Thermocouples/Thermistors:

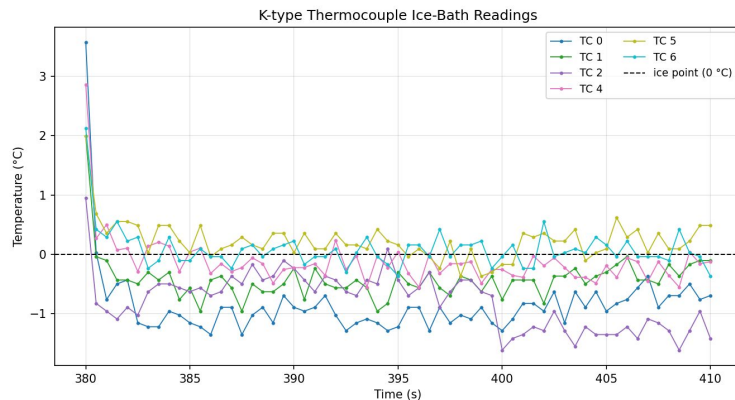
- ice bath calibration 0°C and 100°C water
- $T_{\text{corrected}} = T_{\text{measured}} - \text{offset}$

Thermal Camera:

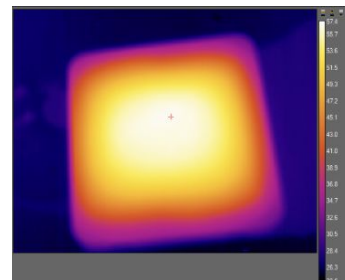
- FLIR AX5 SEQ file stores raw 16-bit integers per pixel, a two-point linear calibration was performed by mapping the raw pixel range to the known temperature range to recover accurate temperatures readings
- $T(^{\circ}\text{C}) = \text{scale} \times S + \text{offset} - 273.15$

Samples/printer:

- Samples printed on same printer (Bambu P1S), identical material, slicing parameters, 100% infill



Channel	Mean (°C)	95% CI (°C)	Range (°C)
TC 0	-0.9	±1.2	[-2.1, +0.4]
TC 1	-0.4	±0.7	[-1.2, +0.3]
TC 2	-0.8	±0.9	[-1.7, +0.2]
TC 4	-0.1	±0.9	[-1.0, +0.8]
TC 5	+0.2	±0.7	[-0.4, +0.9]
TC 6	+0.1	±0.7	[-0.6, +0.7]



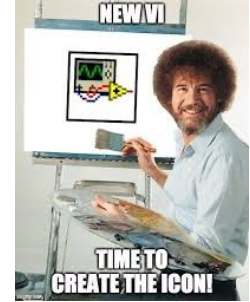
Data collection | Conjugate Heat Transfer

- Sensing: ESP32 Microcontroller, ADC, Thermistors
- Heat gun: “Constant” temperature, fixed nozzle
- Logging: Output from ESP to laptop, recorded in Arduino IDE
- Trial Structure: Each geometry tested at $RD = .15$ & $.30$, plain sleeve ref.
- Uncertainty: Combined uncertainty on ΔT accounts for quantization, calibration offset, and contact resistance error sources



Data collection | Conduction

- Thermocouples: 6 Type K, 3 per lattice
- Signal conditioning: TC signals amplified and compensated pre-DAQ
- DAQ: High resolution analog input module
- LabVIEW VI: DAQ Assistant → scaling + correction → chart → file to TDMS → CSV export
- Hot plate: fixed at set temperature. Confirmed by TC's



DOE 1 | Conjugate Heat Transfer Study

SENSOR JUSTIFICATION & PHYSICAL PRINCIPLES

- NTC thermistors operating range is well above the measurement window
- ADS1115 16-bit ADC gives uncertainty = $\pm 0.002^{\circ}\text{C}$ – negligible
- 2 sensors (1 front, 1 back) allow for averaging, reducing placement random error

DOE JUSTIFICATION

- 8 Conditions: 3 trials each = 24 total runs.
- Two independent variables (geometry, density) enable the geometry comparison.

DOE 2 | Conduction Study

SENSOR JUSTIFICATION & PHYSICAL PRINCIPLES

- 6 Type K thermocouples chosen for compatibility with AD8495 amplifier
- AD8495 amplifier converts TC output to 5 mV/°C analog voltage
- DAQ gives sensitivity equivalent to $\pm 0.002^{\circ}\text{C}$ at 5 mV/°C
- 3 TCs per sample at depths $\frac{1}{4}$, $\frac{1}{2}$, and $\frac{3}{4}$ to capture full gradient

DOE JUSTIFICATION

- 4 geometries $\times$ 3 depths $\times$ 3 trials = 12 total runs with 36 measurements
- $\frac{1}{4}$, $\frac{1}{2}$, and $\frac{3}{4}$ chosen to span sample & capture early steep gradient and near surface saturation

DOE 1 | Conjugate Heat Transfer Study

UNCERTAINTY QUANTITATIVE ABSTRACT SUMMARY

<u>Error Source</u>	<u>Type</u>	<u>Value (°C)</u>
NTC thermistor accuracy	Bias	±0.25
Two-point calibration residual	Bias	±0.15
Thermistor contact/placement	Bias	±0.30
Ambient room temp drift	Random	±0.20*
ADS1115 ADC quantization	Random	±0.002
Thermistor self heating	Random	negligible
Combined known uncertainty (RSS)		±0.4

*Mitigated by normalizing all readings to T₀ at t = 0 **Heat gun uncertainty not included

DOE 2 | Conduction Study

UNCERTAINTY QUANTITATIVE ABSTRACT SUMMARY

<u>Error Source</u>	<u>Type</u>	<u>Value (°C)</u>
AD8495 amplifier accuracy	Bias	±1.00
Type K thermocouple nonlinearity residual	Bias	±0.50
TC insertion depth positioning	Bias	±0.30
Hot plate surface temp non-uniformity	Bias	Mitigated
Ambient room temp drift	Random	±0.20*
TC contact resistance variation between trials	Random	±0.10
DAQ quantization	Random	±0.002
Combined known uncertainty (RSS)		±1.2

*Mitigated by normalizing all readings to T₀ at t = 0

DOE 1 | Conjugate Heat Transfer Study

BIAS ERRORS

- Heat gun output temp not independently measured. Constant temp is an assumption. Unknown estimated bias
- NTC calibration residual after correction: $\pm 0.15^{\circ}\text{C}$ offset
- Contact resistance placement: $\pm 0.10^{\circ}\text{C}$

RANDOM ERRORS

- ADS1115 noise: $\pm 0.002^{\circ}\text{C}$
- Ambient room temperature fluctuation between trials: $\pm 0.5^{\circ}\text{C}$, mitigated by normalization at $t=0$
- Thermistor self-heating: negligible

Combined uncertainty (RSS): $\pm 0.42^{\circ}\text{C}$

Quantitative Abstract: DOE 1

- **DOE 1 - Conjugate Heat Transfer**

- NTC thermistor → voltage divider → ADS1115 16-bit ADC → ESP32
- ADC: 1 LSB = 62.5 μ V, Sensitivity = 40 mV/°C, LSB/Sens = ± 0.0016 °C
- Thermistor: 1% accuracy → At 25°C resistance error → ± 0.25 °C
- Two point calibration: correction eliminates most, using ± 0.15 °C
- Repeatability: Thermistor contact to surface: ± 0.3 °C
- Combined: $u_{\text{total}} = \sqrt{(0.0016^2 + 0.25^2 + 0.15^2 + 0.1^2)} = \underline{\underline{\pm 0.42 \text{ °C}}}$

DOE 2 | Conduction Study

BIAS ERRORS

- Hot plate surface temp non-uniformity: setpoint verified via TC plateau
- Placed samples in same calibrated position
- AD8495 assumes ambient is stable. $\sim 0.05^{\circ}\text{C}/^{\circ}\text{C}$
- TC insertion depth: repeatable to ± 0.5 mm

RANDOM ERRORS

- TC contact resistance variation between trials: $\pm 0.10^{\circ}\text{C}$
- Hot plate temp oscillation: $\pm 0.2^{\circ}\text{C}$ random variation
- Ambient temp fluctuation $\pm 0.5^{\circ}\text{C}$

Combined uncertainty (RSS): $\pm 1.16^{\circ}\text{C}$

Quantitative Abstract: DOE 2

- **DOE 2 - Conduction**

- Type K TC → AD8495 amplifier → NI USB-6211 DAQ
- DAQ: Sensitivity = 10.4 μV , AD sensitivity = 5 mV/ $^{\circ}\text{C}$ → ± 0.002 $^{\circ}\text{C}$
- AD8495: $\pm 1^{\circ}\text{C}$ at room temperature
- Type K thermocouple: Standard: $\pm 2.5^{\circ}\text{C}/\pm 0.75\%$, captured in AD, $\pm 0.5^{\circ}\text{C}$
- Repeatability: Thermistor contact to surface: ± 0.1 $^{\circ}\text{C}$
- Combined: $u_{\text{total}} = \sqrt{(0.002^2 + 1.0^2 + 0.5^2 + 0.1^2)} = \underline{\underline{\pm 1.16}}^{\circ}\text{C}$

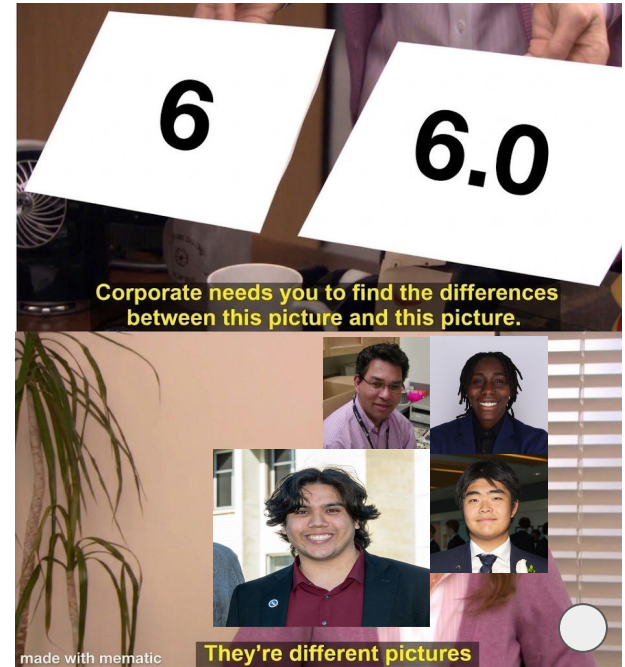
DOE 1 | Conjugate Heat Transfer Study

IDEALIZED MODEL ASSUMPTIONS

- Heat gun treated as uniform, steady state heat flux source...
- Sample surfaces treated as isothermal planes
- No heat loss from sample sides
- PLA thermal properties assumed uniform + consistent

SIG FIGS

- Temp reported 0.1°C , consistent with combined uncertainty of $\pm 0.4^{\circ}\text{C}$. ΔT reported to 0.1°C .



DOE 2 | Conduction Study

IDEALIZED MODEL ASSUMPTIONS

- Hot plate bottom face treated as uniform & constant temp boundary condition
- Treated as 1D conduction problem
- PLA conductivity treated is isotropic & uniform
- Perfect thermal constant assumed between base & hot plate surface

SIG FIGS

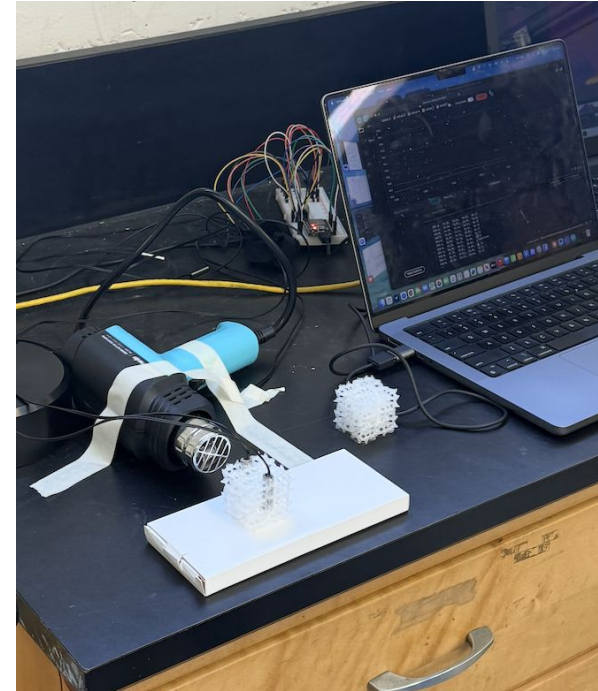
- Temp reported 0.1°C , consistent with combined uncertainty of $\pm 1.2^{\circ}\text{C}$. Steady state values reported to 1 decimal places



DOE 1 | Conjugate Heat Transfer Study

PROCEDURE

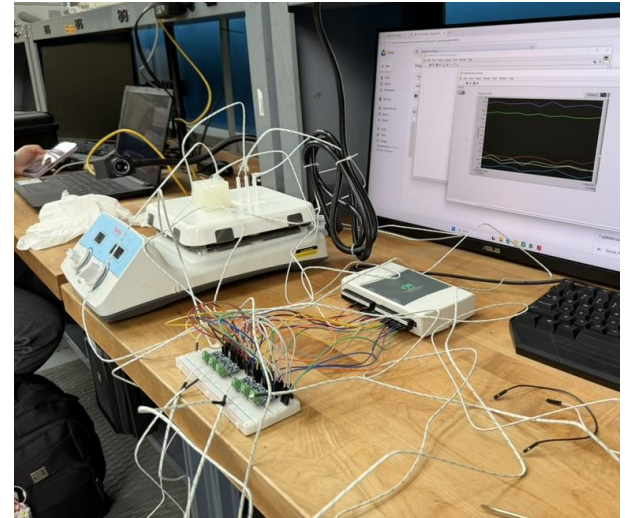
1. Insert one thermistor $\frac{1}{4}$ from front face, and one $\frac{3}{4}$ from front face
2. Position heat gun nozzle at a fixed distance, let it reach steady temp
3. Begin serial logging in Arduino IDE, confirm ambient temp
4. Turn on heat gun, record both channels for 300s at 1Hz
5. Export serial log to csv, repeat for all other geometries



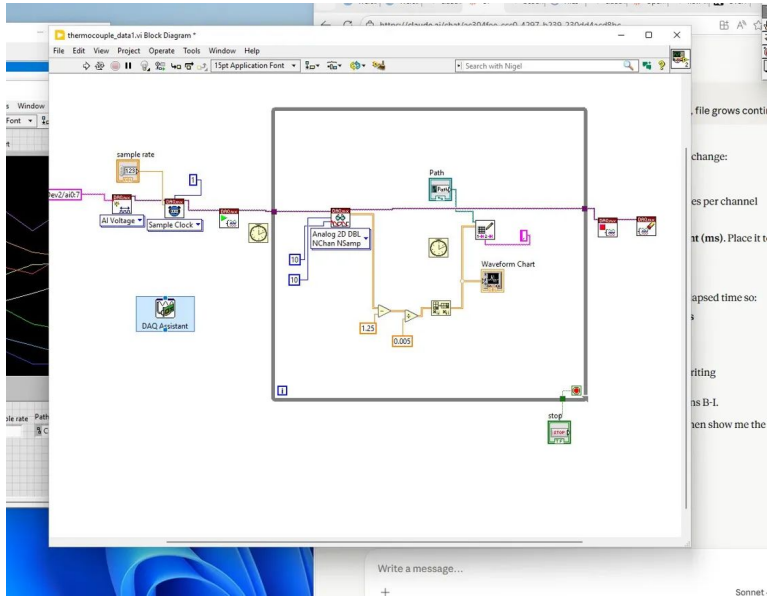
DOE 2 | Conduction Study

PROCEDURE

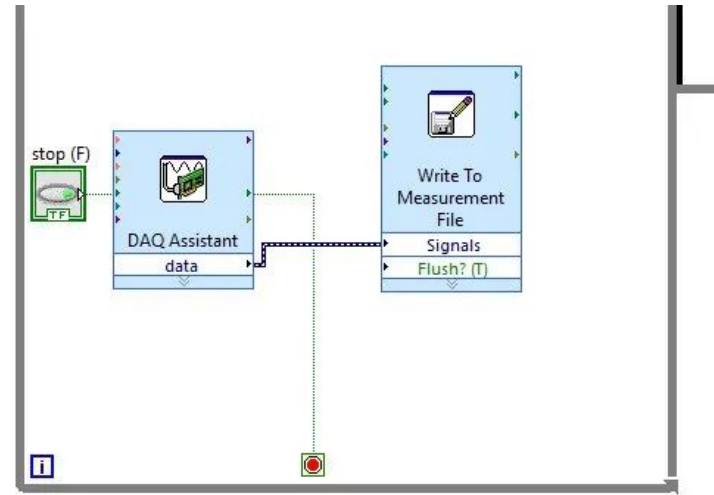
1. Insert 3 Type K TCs per sample at $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, depth positions, route via AD8495 to DAQ
2. Set hot plate to target temp, wait for TC readings to plateau
3. Place both samples simultaneously on the hot plate, start LabVIEW VI
4. LabVIEW flow: DAQ Assistant → linear scaling + calibration correction → chart → export CSV
5. Log continuously until all channels reach steady state defined as $\Delta T/\Delta t < 0.05^{\circ}\text{C}/\text{min}$
6. Remove samples, allow samples return to ambient before next trial



DAQ Journey...

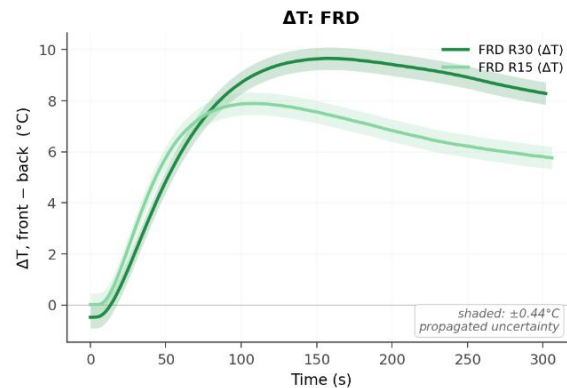
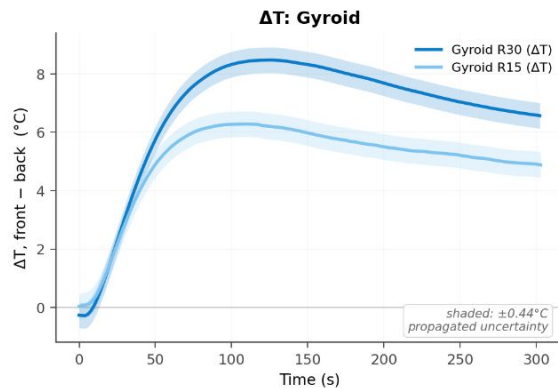
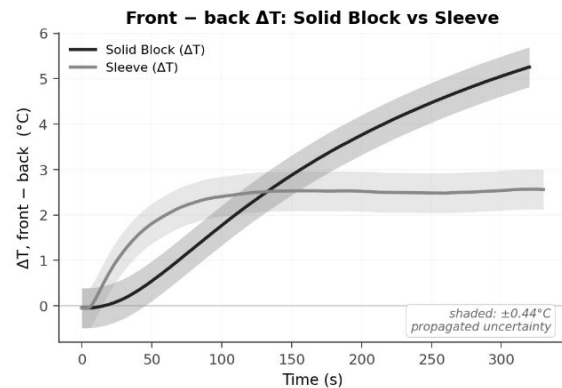
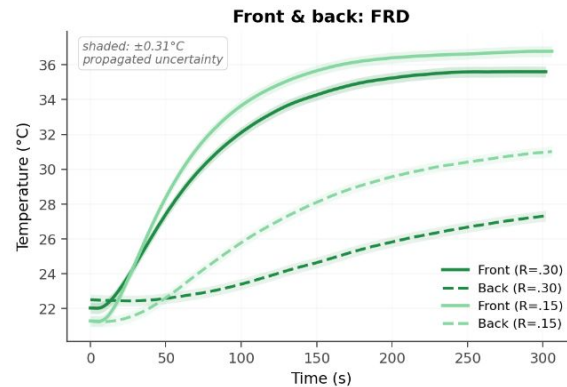
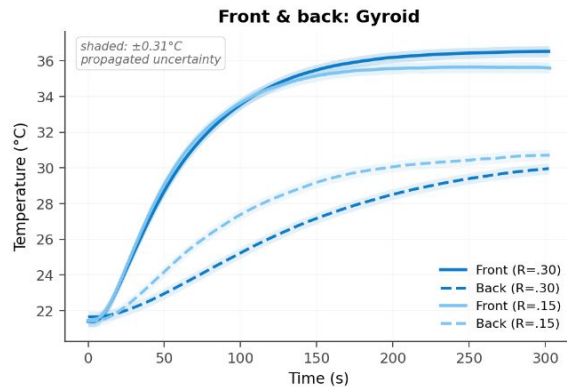
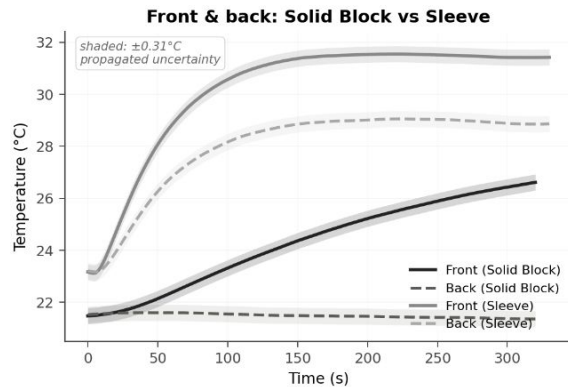


Before DAQ Assistant 1 (1 BDA)

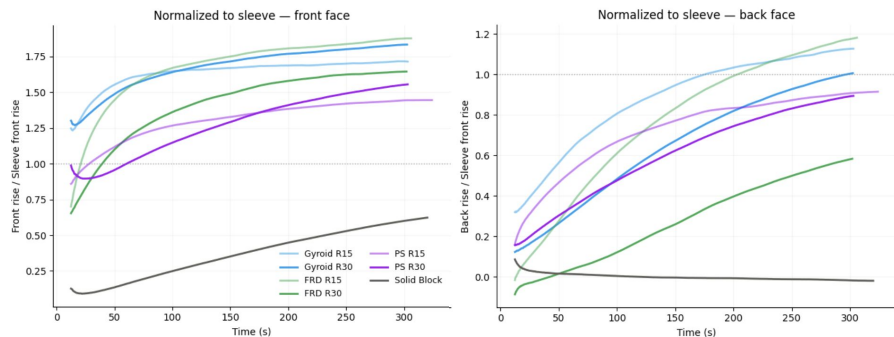


After DAQ Assistant 1 (1 ADA)

Results | Conjugate Heat Transfer

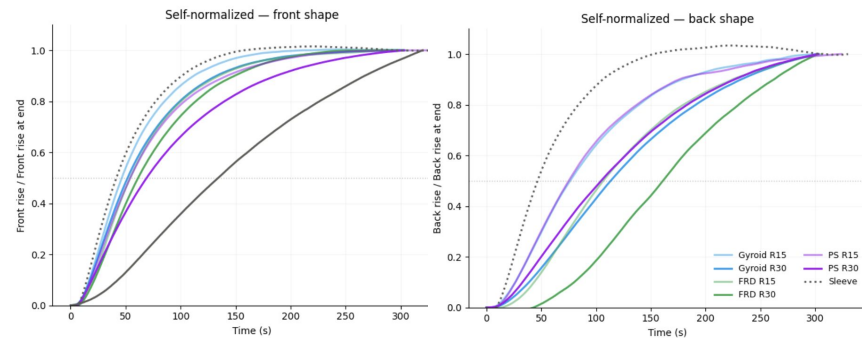


Results | Conjugate Heat Transfer



$$y_{\text{front}}(t) = \frac{F(t)}{F_{\text{sleeve}}(t)}, \quad y_{\text{back}}(t) = \frac{B(t)}{F_{\text{sleeve}}(t)}$$

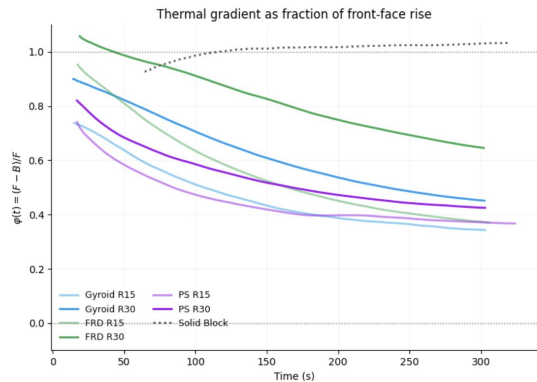
- Removes heat-gun variability by dividing through by the empty-rig response
- → Only FRD R30 is meaningfully blocking



$$y_{\text{front}}(t) = \frac{F(t)}{F_{\infty}}, \quad y_{\text{back}}(t) = \frac{B(t)}{B_{\infty}}$$

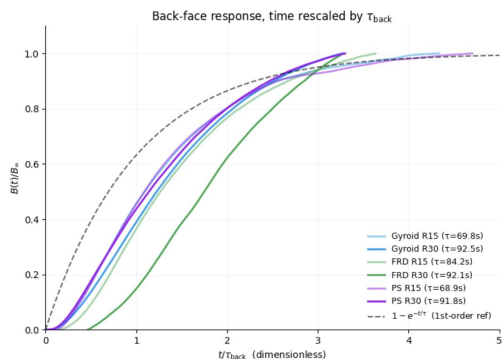
- Divides each curve by its own final value
- $y=0.5$ crossing = half-rise time.
FRD R30 crosses at ~175 s vs ~100–120 s for others
- → FRD R30 is genuinely slower and blocks more both help

Results | Conjugate Heat Transfer



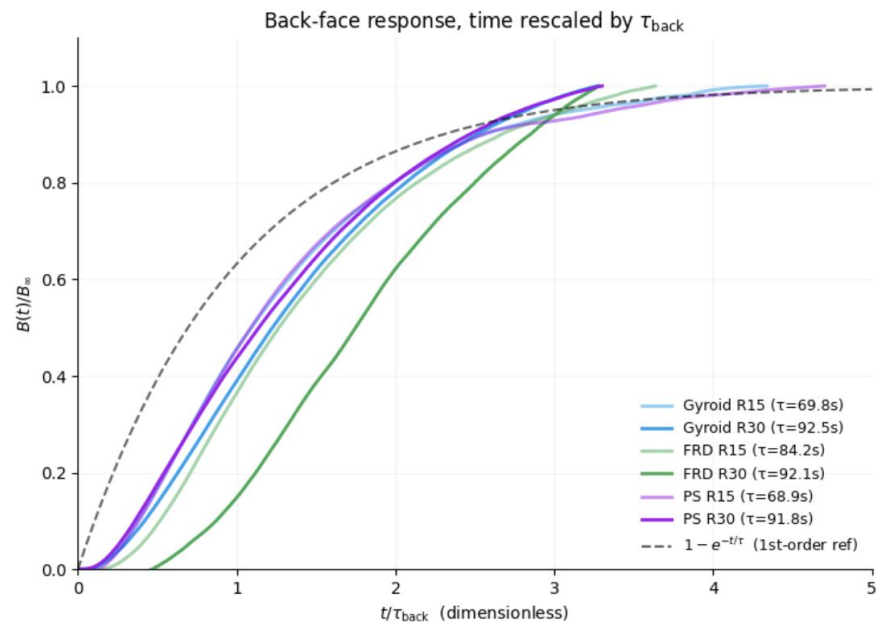
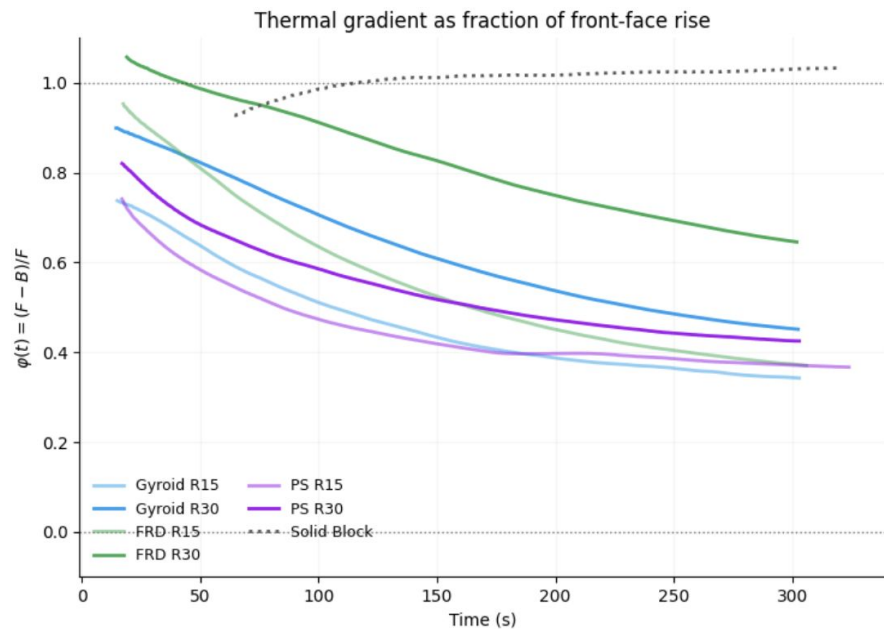
$$\varphi(t) = \frac{F(t) - B(t)}{F(t)}$$

- Bounded $[0, 1]$: 1 = perfect blocker, 0 = transparent
- → FRD R30 maintains a real spatial gradient; others equalize

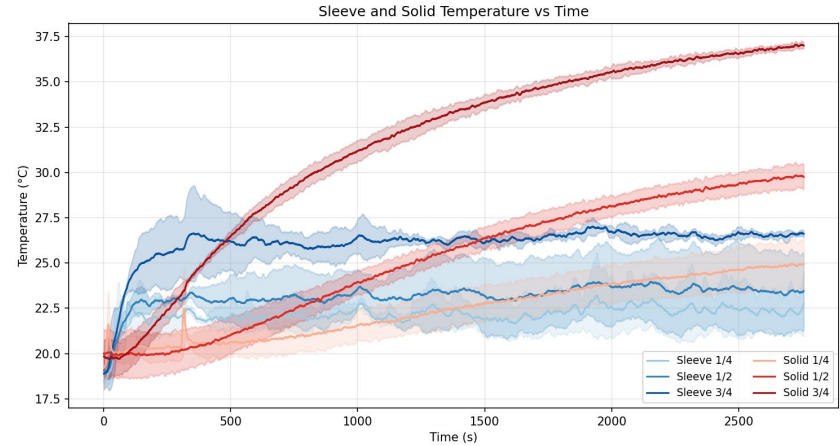
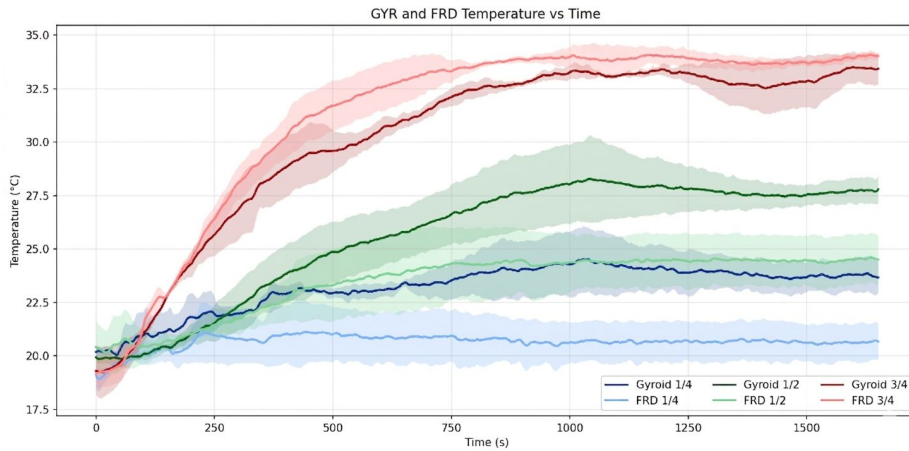


$$B(t) = B_{\infty} \left(1 - e^{-t/\tau}\right)$$

- Both axes dimensionless, dashed reference = ideal first-order system
- → FRD R30 seems to block by conduction, not flow obstruction



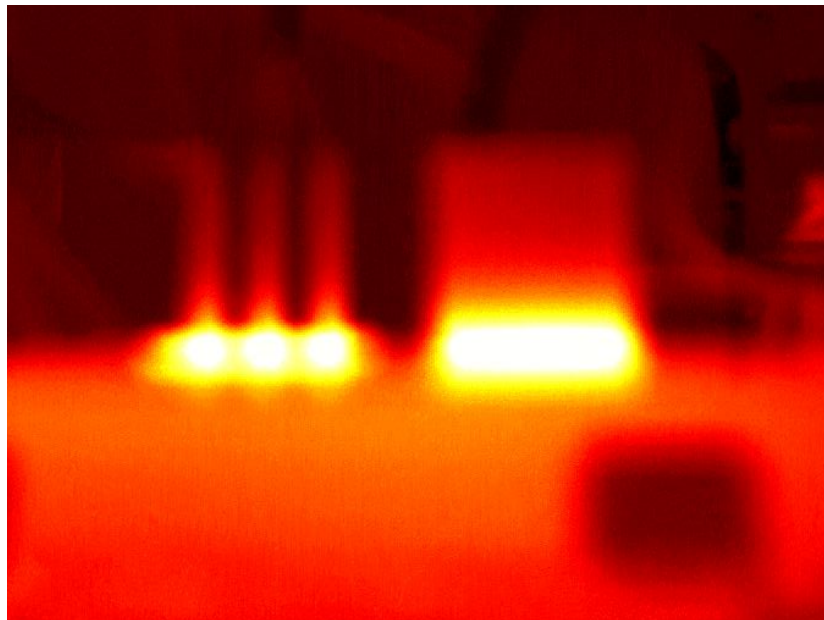
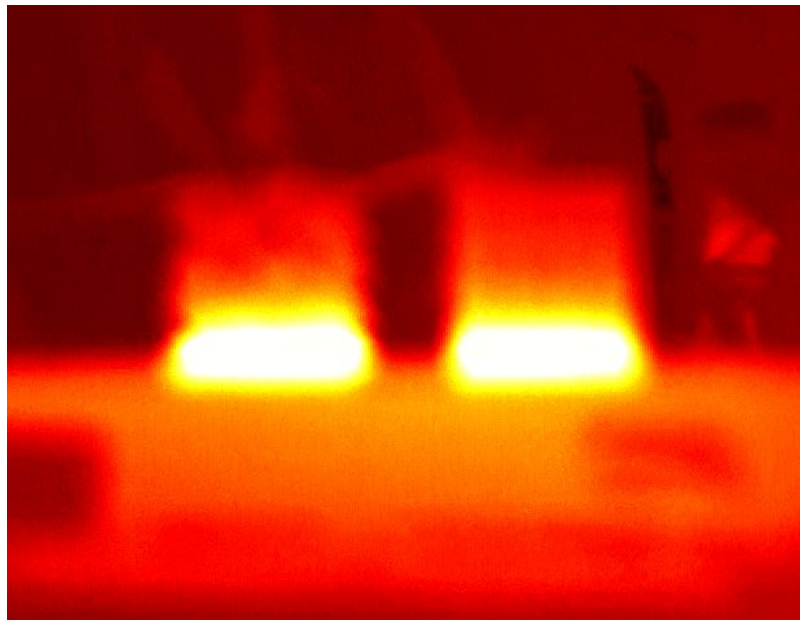
Results | Conduction



- FRD heats up faster and reaches higher temperatures at every depth than Gyroid
- Gyroid actually conducts heat upward better than FRD
- solid control is repeatable; both trials within ~1 °C, confirms the heat input was consistent
- Lattice samples show overshoot, control doesn't

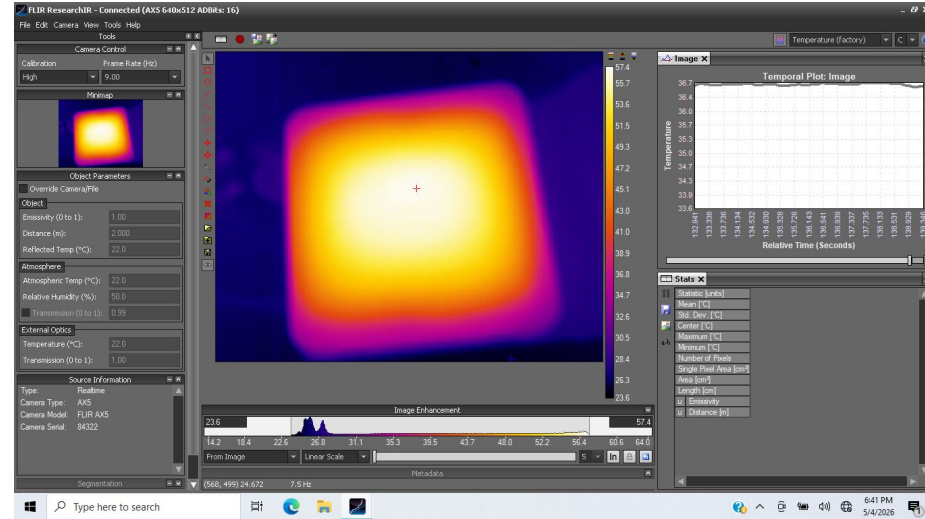
→ Data suggests convective heat transfer through open lattice channels and null control, not just conduction

Data Processing | Thermal Camera



Thermal Camera FLIR AX5 Calibration

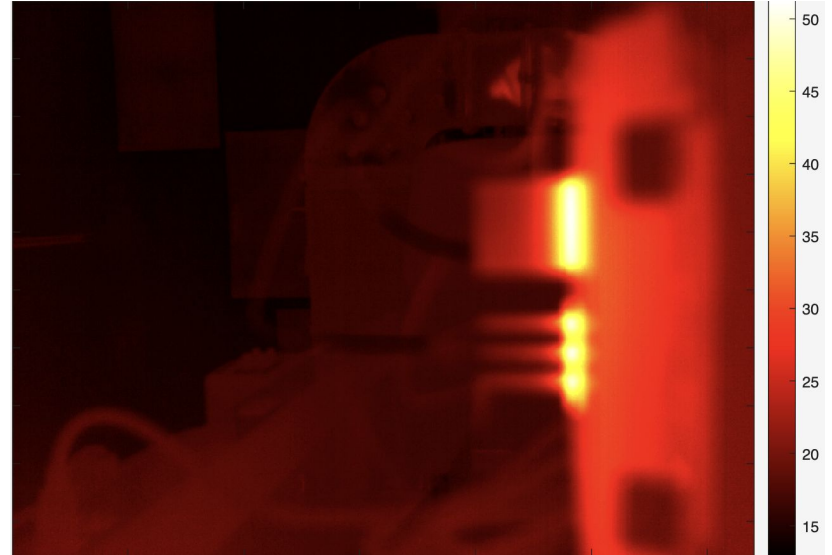
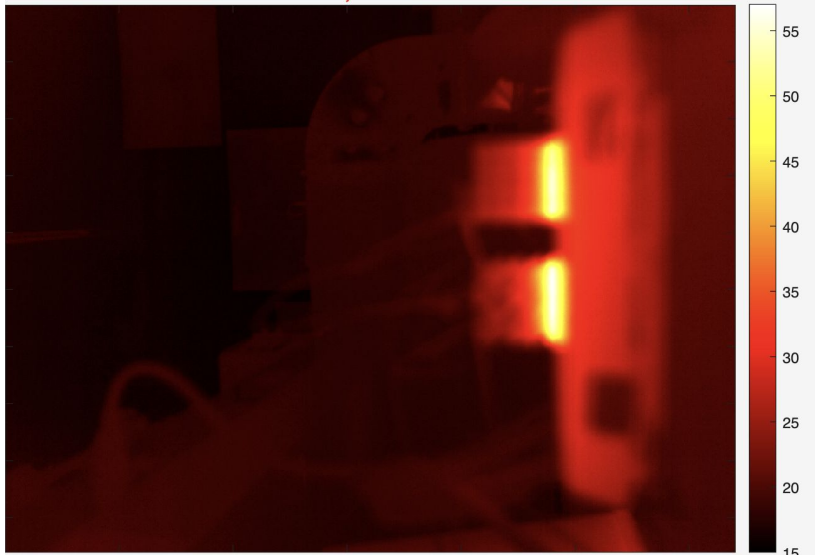
- SEQ file store data as raw 16-bit integers
 - Conversion methods: Planck/Radiometric formula, Temperature-linear formula
- Turns out to be: $T(^{\circ}\text{C}) = \text{scale} \times S + \text{offset} - 273.15$
- Find the scale and offset (S is the raw 16-bit value from one pixel)
 - $\text{scale} = (T_{\text{hot}} - T_{\text{cold}}) / (S_{\text{hot}} - S_{\text{cold}}) = 0.026415 \text{ K/count}$, s is the raw count
 - $\text{offset} = (55 + 273.15) - \text{scale} \times S = 208.3236 \text{ K}$
- Converts every pixel of every frame to the correct temperature value!



FLIR AX5 Calibration Results

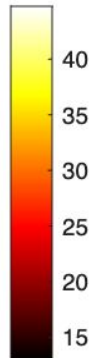
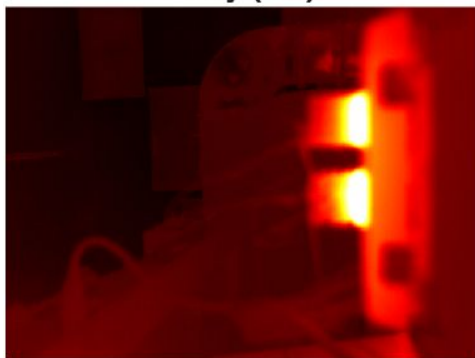
FOR ALL 4 GEOMETRIES

- Calibrated thermal camera used to verify thermocouple data

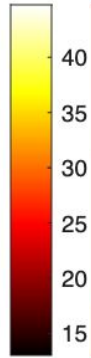
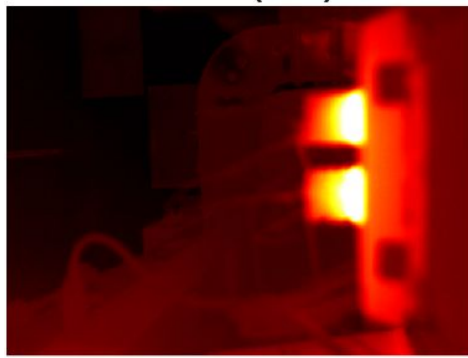


Thermal Camera: Results

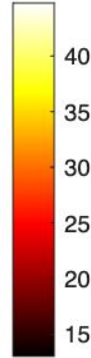
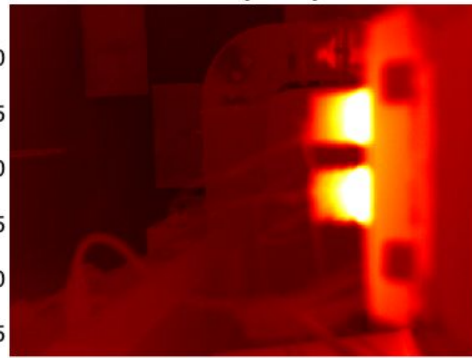
Early (5%)



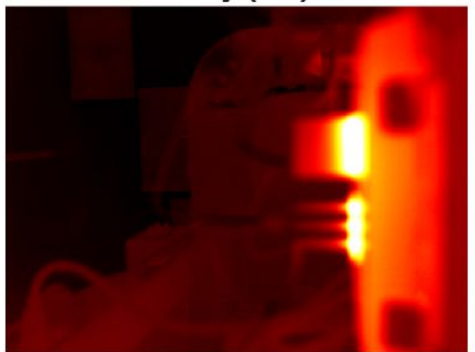
Middle (45%)



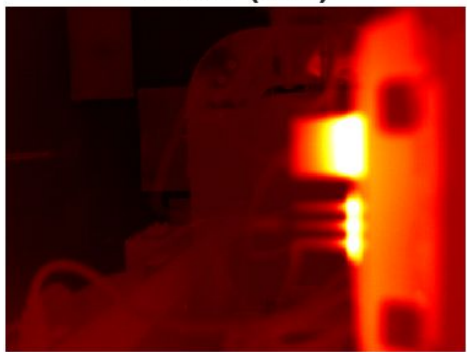
Late (90%)



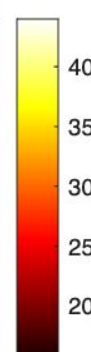
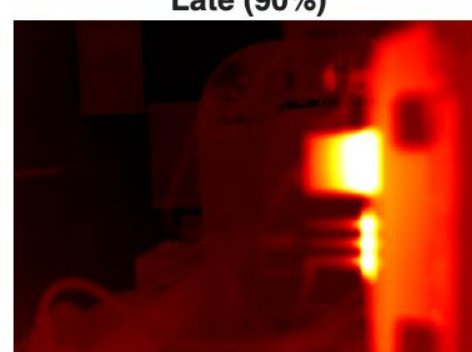
Early (5%)



Middle (45%)

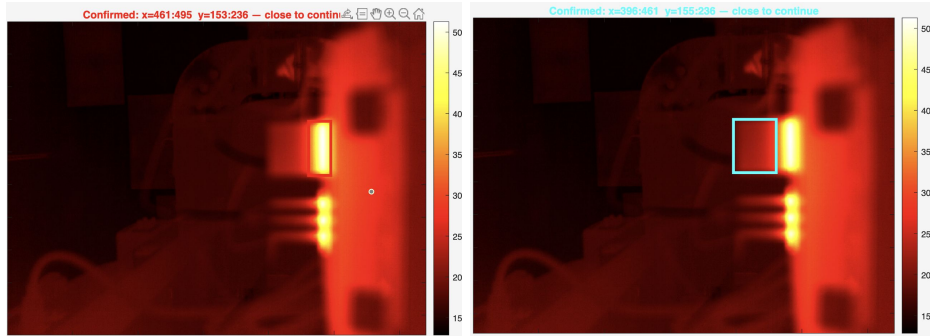


Late (90%)

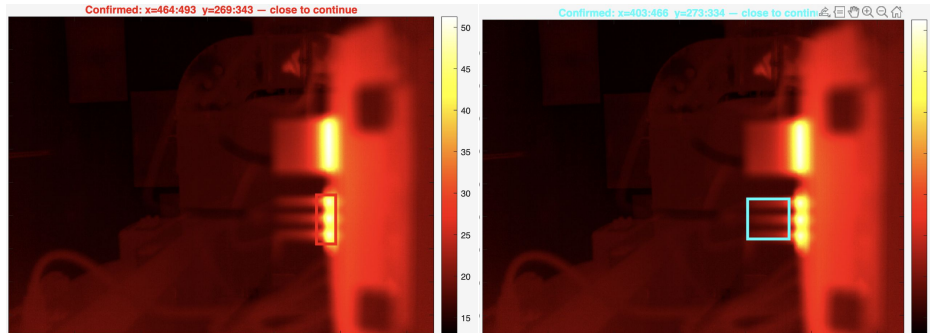


Thermal Camera: Selections

SOLID



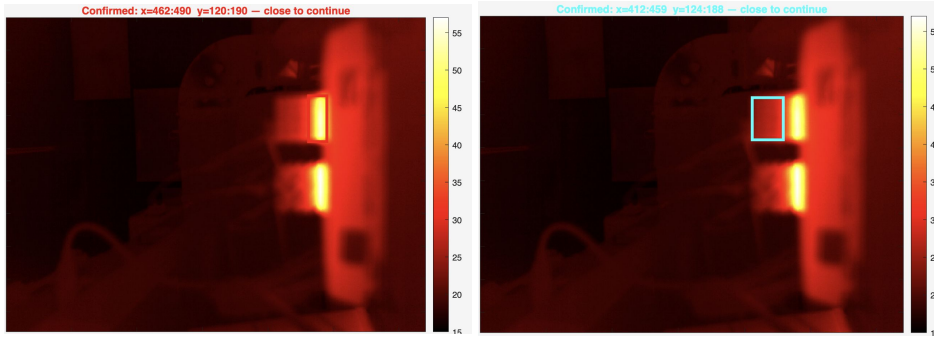
SLEEVE



- Solid block: low SA:V, heat is conducted uniformly with minimal convective
- The sleeve: medium SA:V, reduced material cross-section limits conductive pathways, the gaps allow convective air flow

Thermal Camera: Selections

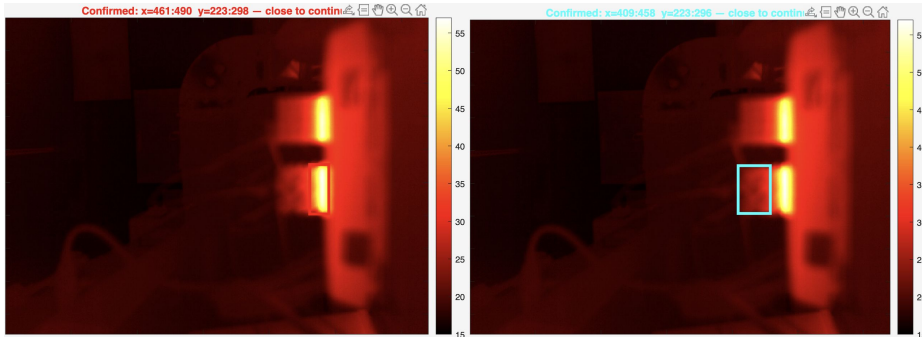
FRD



Small ΔT = good thermal conductor geometry

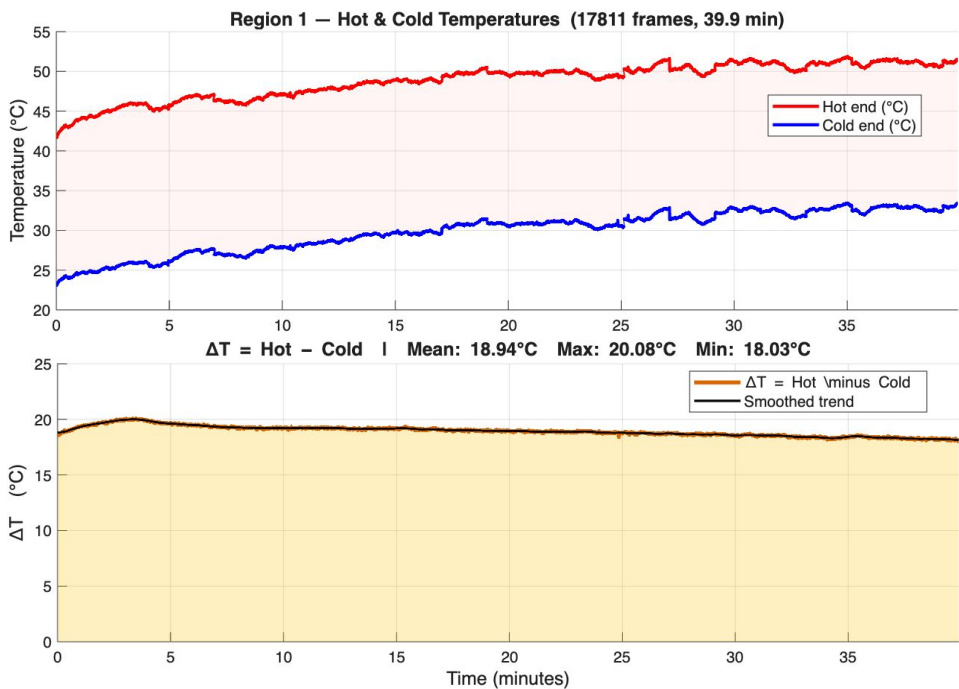
Large ΔT = good thermal insulator geometry

GYROID

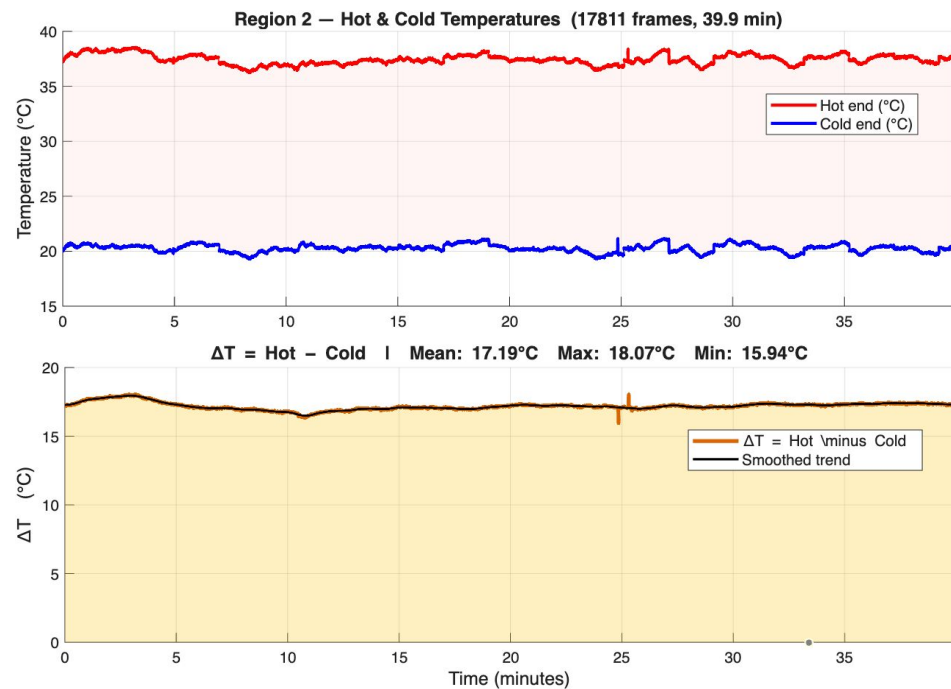


Thermal Camera: Results

SOLID

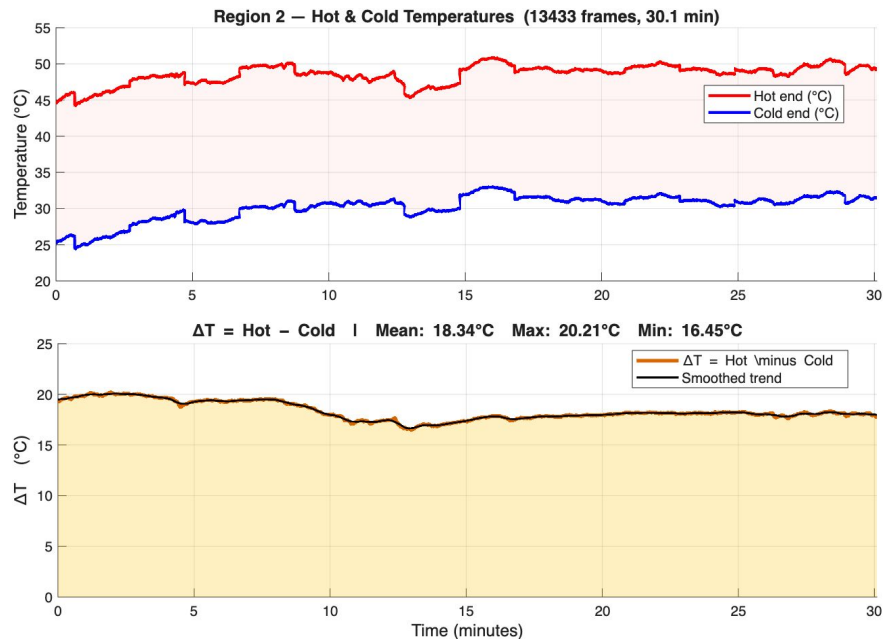


SLEEVE

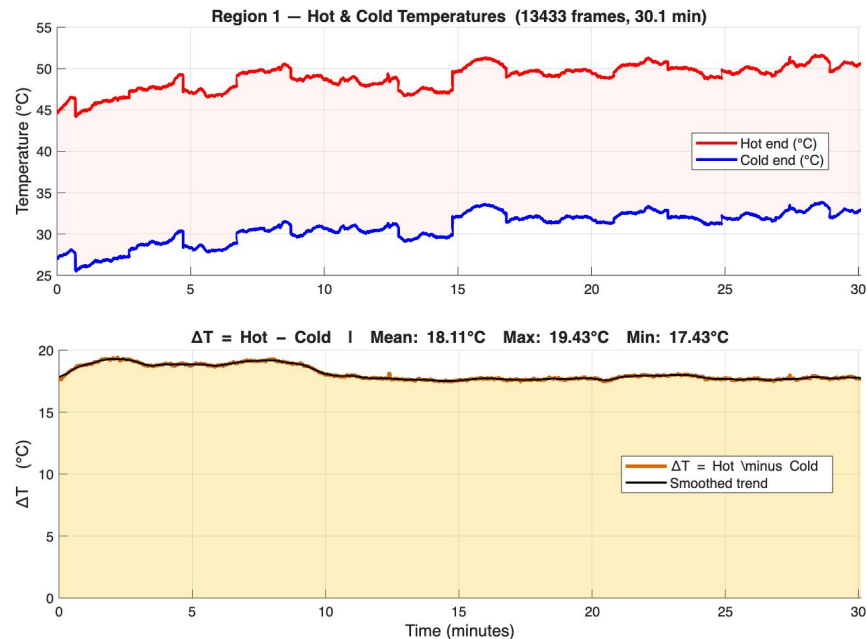


Thermal Camera: Results

GYROID

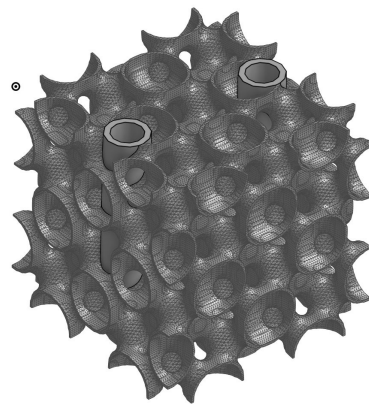


FRD



Conclusions & Key Findings

- FRD R30 is the only lattice that actually blocks heat: its back face stayed 40% cooler
- FRD blocks heat by being a bad conductor, not by disrupting airflow: the back-face curve shows heat taking time to travel through the solid struts to reach the back.
- Density only helps if the geometry is tortuous to begin with: bumping Gyroid and PS from 15% to 30% density changed almost nothing.



Improvements on the Experimentations

Replace heat gun with flow-controlled convection source

Embed sensors in the printed channels to eliminate contact resistance error

Systematic RD range (0.10–0.50, step 0.05) to fully showcase density performance relationship

Print in metal for realistic thermal conductivity

Increase to $n \geq 5$ trials per condition for statistical confidence intervals

CFD validation to decouple convective vs. conductive contributions

Lessons Learned & Future Studies

- Flow control limitations
- Sensor Placement consistency
- Print quality & material variability
- Convection → conduction pivot



- Markforged Metal X is questionable
- Systemic density sweep
- CFD validation via simulation

References

1. Bergman et al. (2017). Fundamentals of Heat and Mass Transfer (8th ed.). Wiley.
2. Al-Ketan et al. (2018). Topology-mechanical property relationship of 3D printed TMPS lattices. Additive Manufacturing, 19, 167–183.

Hypothesis Test Results

Thermal Response of
TMPS Lattice Structures

Sonali Das, Daniel Roberts, Tianning Huang



DOE 1 | Conjugate Heat Transfer

HYPOTHESIS

- H_0 : Lattice topology has no effect on through-thickness heat transfer: all geometries produce equivalent front-to-back ΔT responses at fixed relative density
- H_1 : Topology produces measurably different heat-transfer responses

TEST: TWO-SAMPLE T-TEST

- Two-tailed, $\alpha = 0.05$, $n = 3$ trials per condition
- Decision rule: Reject H_0 if $|t| > t_{\text{crit}}$, where $t_{\text{crit}} = t_{0.975, v}$

DOE 1 | Conjugate Heat Transfer

TIME DEPENDENCE OF EXPERIMENTAL DATA

- 3 samples per geometry × 2 relative densities (RD = 0.15, 0.30)
- ↓
- Two-sample, two-tailed t-test

$$t = \frac{\bar{x}_1 - \bar{x}_2}{SE}, \quad SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

DOE 1 | Conjugate Heat Transfer

STEADY STATE ΔT | RAW DATA SUMMARY (n=3)

ΔT = front thermistor rise minus back thermistor rise. Last 25% of 300 s recording minus 20-sample baseline.

Condition	Trial 1 (°C)	Trial 2 (°C)	Trial 3 (°C)	$\bar{x}$ (°C)	s (°C)
FRD R30	9.166	7.794	8.225	8.395	0.702
FRD R15	6.025	4.283	7.382	5.897	1.554
GYR R30	7.122	7.831	8.088	7.68	0.5
GYR R15	4.999	5.079	5.39	5.156	0.206
PS R30	5.458	5.094	7.577	6.043	1.341
PS R15	4.491	3.781	3.385	3.886	0.56
Solid	4.722	6.12	1.646	4.163	2.289
Sleeve	2.535	4.404	4.475	3.805	1.1

- Larger ΔT = steeper gradient = geometry resists heat flow more
- R30 conditions consistently outperform their R15 counterparts within every topology
- FRD R30 and GYR R30 are the two highest performers while Sleeve and PS R15 are the lowest

DOE 1 | Conjugate Heat Transfer

T-TEST RESULTS | ALL 28 PAIRWISE COMPARISONS

Two-sample t-test, two-tailed, $\alpha = 0.05$. Bold = Reject H_0 . Unbold = Fail to reject.

Comparison	$\bar{x}_1$	$\bar{x}_2$	SE	t	v/t_crit	p	Decision
FRD R30 vs. GYR R30	8.395	7.68	0.498	1.44	3.62/2.897	0.2314	Fail
FRD R30 vs. PS R30	8.395	6.043	0.874	2.69	3.02/3.171	0.0738	Borderline
FRD R30 vs. Solid	8.395	4.163	1.382	3.06	2.37/3.716	0.0744	Borderline
FRD R30 vs. Sleeve	8.395	3.805	0.753	6.09	3.40/2.982	0.0061	✓ Reject
FRD R30 vs. FRD R15	8.395	5.897	0.965	2.59	2.19/3.940	0.1143	Fail
FRD R30 vs. GYR R15	8.395	5.156	0.735	4.41	2.10/4.137	0.0429	✓ Reject
FRD R30 vs. PS R15	8.395	3.886	0.729	6.18	2.77/3.321	0.0093	✓ Reject
GYR R30 vs. PS R30	7.68	6.043	0.841	1.95	2.56/3.519	0.1649	Fail
GYR R30 vs. Solid	7.68	4.163	1.347	2.61	2.27/3.840	0.1122	Fail
GYR R30 vs. Sleeve	7.68	3.805	0.661	5.86	2.32/3.777	0.0148	✓ Reject
GYR R30 vs. FRD R15	7.68	5.897	0.923	1.93	2.15/4.063	0.184	Fail
GYR R30 vs. GYR R15	7.68	5.156	0.293	8.62	2.04/4.161	0.0126	✓ Reject
GYR R30 vs. PS R15	7.68	3.886	0.686	5.53	2.56/3.519	0.0195	✓ Reject
PS R30 vs. Solid	6.043	4.163	1.654	1.14	2.07/4.161	0.3722	Fail

Comparison	$\bar{x}_1$	$\bar{x}_2$	SE	t	v/t_crit	p	Decision
PS R30 vs. Sleeve	6.043	3.805	1.394	1.61	2.07/4.168	0.2488	Fail
PS R30 vs. FRD R15	6.043	5.897	1.588	0.09	2.09/4.145	0.9318	Fail
PS R30 vs. GYR R15	6.043	5.156	1.351	0.66	2.09/4.145	0.5877	Fail
PS R30 vs. PS R15	6.043	3.886	1.379	1.56	2.08/4.153	0.2577	Fail
Solid vs. Sleeve	4.163	3.805	2.113	0.17	2.37/3.716	0.8784	Fail
Solid vs. FRD R15	4.163	5.897	2.023	-0.86	2.34/3.747	0.4703	Fail
Solid vs. GYR R15	4.163	5.156	2.295	-0.43	2.37/3.714	0.7127	Fail
Solid vs. PS R15	4.163	3.886	2.298	0.12	2.37/3.714	0.9119	Fail
Sleeve vs. FRD R15	3.805	5.897	1.396	-1.5	2.08/4.156	0.2733	Fail
Sleeve vs. GYR R15	3.805	5.156	1.122	-1.2	2.12/4.103	0.3488	Fail
Sleeve vs. PS R15	3.805	3.886	1.149	-0.07	2.29/3.822	0.948	Fail
FRD R15 vs. GYR R15	5.897	5.156	0.905	0.82	2.07/4.165	0.4965	Fail
FRD R15 vs. PS R15	5.897	3.886	0.953	2.11	2.51/3.563	0.1429	Fail
GYR R15 vs. PS R15	5.156	3.886	0.594	2.14	2.56/3.519	0.1501	Fail

DOE 1 | Conjugate Heat Transfer

T-TEST RESULTS | ALL 28 PAIRWISE COMPARISONS

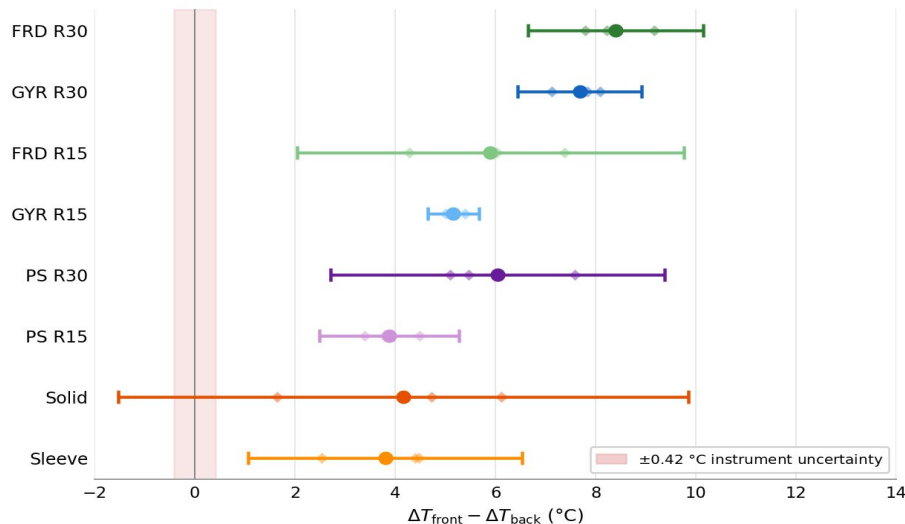
Two-sample t-test, two-tailed, $\alpha = 0.05$. Bold = Reject H_0 . Unbold = Fail to reject.

- Full p-value bar chart across all 28 comparisons, grouped
- Visually shows that all rejections cluster in the R30 vs. low-performer region

DOE 1 | Conjugate Heat Transfer

VISUALIZATIONS | MEAN ΔT WITH 95% CONFIDENCE INTERVALS

DOE 1 Conjugate Heat Transfer — Mean ΔT with 95% Confidence Intervals
(Welch t -distribution, $n = 3$; shaded band = ± 0.42 °C instrument uncertainty)



- R30 conditions sit clearly right of R15 conditions: density effect visible
- FRD R15 and Solid have the widest CIs: need additional trials
- GYR R15 has the tightest CI: three trials were unusually consistent
- No R30 CIs are separated from each other, substantial overlap
- Sleeve and PS R15 sit left: conditions most distinguished from the R30
- Some conditions have CIs that approach or cross zero relative to the instrument noise floor

Each point is the group mean; error bars are 95% CI (t-distribution, $n=3$); diamonds are individual trial values; pink band = ± 0.42 °C instrument uncertainty

DOE 1 | Conjugate Heat Transfer

STATISTICAL CONCLUSIONS | HYPOTHESIS CONCLUSION

Verdict: 5 of 28 pairwise comparisons reject H_0 at $\alpha = 0.05$. All rejections involve R30-density vs. lower-performing conditions. No within-density-tier comparison is significant.

What the data does show:

- Density is the primary driver: topology alone cannot explain the pattern
- Topology is not resolvable at $n=3$: FRD R30 vs. GYR R30 ($p = 0.231$), FRD R15 vs. GYR R15 ($p = 0.497$), none reject H_0
- FRD R30 vs. Sleeve is the strongest result ($p = 0.006$, $|t| = 6.09$) and the most physically meaningful given Sleeve's role as the empty reference

DOE 2 | Conduction Study

HYPOTHESIS

- H_0 : Lattice topology has no effect on through-thickness heat transfer. All geometries produce equivalent steady-state temperature gradients at fixed relative density
- H_1 : Topology produces measurably different heat-transfer responses

TEST: TWO-SAMPLE T-TEST

- Two-tailed, $\alpha = 0.05$, $n = 3$ trials per condition
- Decision rule: Reject H_0 if $|t| > t_{\text{crit}}$, where $t_{\text{crit}} = t_{0.975, \nu}$

DOE 1 | Conduction Study

TIME DEPENDENCE OF EXPERIMENTAL DATA

- 3 samples per geometry × 3 depth positions ($\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$)
- ↓
- Two-sample, two-tailed t-test

$$t = \frac{\bar{x}_1 - \bar{x}_2}{SE}, \quad SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

DOE 2 | Conduction Study

STEADY STATE ΔT | RAW DATA SUMMARY (n=3)

All values are mean temperature rise from ambient ($^{\circ}\text{C}$) at steady state, last 25% of recording minus 20-sample baseline

<u>Geometry</u>	<u>$\frac{1}{4}$ depth- top</u> <u>(ambient side)</u>	<u>$\frac{1}{2}$ depth- middle</u>	<u>$\frac{3}{4}$ depth- bottom</u> <u>(hot plate side)</u>
Gyroid	4.3 $\pm$ 1.5	7.7 $\pm$ 0.9	14.0 $\pm$
FRD	2.4 $\pm$ 1.9	3.5 $\pm$ 2.5	14.4 $\pm$ 0.5
Sleeve	3.9 $\pm$ 0.9	5.0 $\pm$ 1.5	9.2 $\pm$ 2.8
Solid	5.2 $\pm$ 0.3	8.9 $\pm$ 1.0	15.3 $\pm$ 2.8

- The heat gradient runs bottom $\rightarrow$ top
- Steep gradient = heat is not propagating upward = geometry resists conduction.
- A flat gradient = heat equalizes easily = geometry conducts well.

DOE 2 | Conduction Study

T-TEST RESULTS | HOT PLATE SIDE (³/₄ DEPTH)

³/₄ depth (top)

Comparison	$\bar{x}_1$ (°C)	$\bar{x}_2$ (°C)	$\Delta\bar{x}$	t	v	p	Result
FRD vs Gyroid	14.44	14	0.44	1.02	3.99	0.366	Fail
FRD vs Sleeve	14.44	9.19	5.25	3.24	2.16	0.076	Fail
FRD vs Solid	14.44	15.25	-0.81	-0.5	2.16	0.662	Fail
Gyroid vs Sleeve	14	9.19	4.81	2.97	2.14	0.09	Fail
Gyroid vs Solid	14	15.25	-1.25	-0.78	2.14	0.513	Fail
Sleeve vs Solid	9.19	15.25	-6.06	-2.7	4	0.054	Fail

- Multiple comparisons all fall just above $\alpha=0.05$. The physical differences of 5–6°C far exceed the $\pm 1.16^\circ\text{C}$ instrument uncertainty. The test lacks power at $n=3$.
- FRD = Gyroid: $p=0.366$ statistically indistinguishable in through thickness conduction at these densities.

DOE 2 | Conduction Study

T-TEST RESULTS | HOT PLATE SIDE ($\frac{1}{2}$ DEPTH)

$\frac{1}{2}$ depth (middle)

Comparison ▾	$\Delta\bar{x}$ (°C) ▾	t ▾	p ▾	Result ▾
FRD vs Gyroid	-4.14	-2.67	0.093	Fail
FRD vs Sleeve	-1.45	-0.86	0.451	Fail
FRD vs Solid	-5.24	-3.33	0.055	Fail
Gyroid vs Sleeve	2.69	2.73	0.064	Fail
Gyroid vs Solid	-1.1	-1.41	0.232	Fail
Sleeve vs Solid	-3.79	-3.7	0.025	✓ Reject

- The only rejection in the entire study: Sleeve transmits less heat to mid-depth than Solid ($p=0.025$).
- Topology effects strengthen with distance from heat source – differences are seen at $\frac{3}{4}$ and $\frac{1}{2}$ depth but not at $\frac{1}{4}$ depth, where temperature rises are small.

DOE 2 | Conduction Study

T-TEST RESULTS | HOT PLATE SIDE ($\frac{1}{4}$ DEPTH)

$\frac{1}{4}$ depth (ambient/top side)

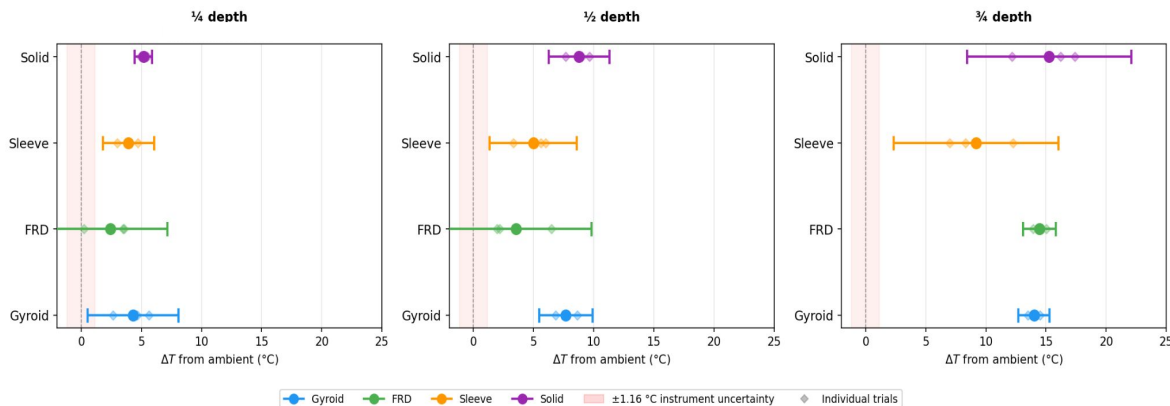
Comparison ▾	$\Delta\bar{x}$ (°C) ▾	t ▾	p ▾	Result ▾
FRD vs Gyroid	-1.72	-1.04	0.409	Fail
FRD vs Sleeve	-1.43	-1.25	0.306	Fail
FRD vs Solid	-2.88	-2.11	0.277	Fail
Gyroid vs Sleeve	0.29	0.17	0.884	Fail
Gyroid vs Solid	-1.16	-0.77	0.58	Fail
Sleeve vs Solid	-1.45	-1.68	0.325	Fail

- All 6 fail to reject H_0
- Signals are small relative to uncertainty – group means range only 2.4–5.2°C against $\pm 1.16^\circ\text{C}$ instrument uncertainty
- Heat hasn't fully propagated to the ambient side within the measurement window

DOE 2 | Conduction Study

VISUALIZATIONS | MEAN ΔT WITH 95% CONFIDENCE INTERVALS

DOE 2 Conduction — Mean ΔT with 95% Confidence Intervals
(t-distribution, $n=3$; shaded band = $\pm 1.16^\circ\text{C}$ instrument uncertainty)

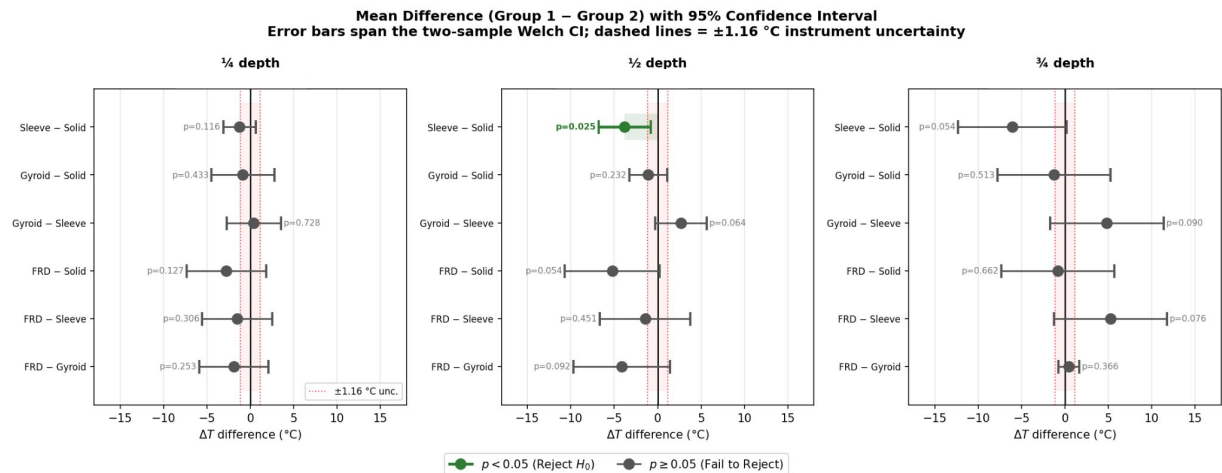


- 1/4: All CIs overlap, no separation at hot plate side
- 1/2: Sleeve separates from Solid
- 3/4: Sleeve CI sits lower but overlaps at $n=3$, wide CIs on Sleeve and Solid reflect variability throughout the geometry.

Each point is the group mean; bars are 95% confidence intervals (t-distribution, $n=3$); diamonds are individual trial values; red band is $\pm 1.16^\circ\text{C}$ instrument uncertainty.

DOE 2 | Conduction Study

VISUALIZATIONS | MEAN ΔT WITH 95% CONFIDENCE INTERVALS



- The only rejection is at $\frac{1}{2}$ depth – Sleeve vs Solid ($p=0.025$, green)
- The Sleeve comparisons show 5–6 $^\circ\text{C}$ gaps, but the sample size is too small to confirm the differences
- FRD and Gyroid are interchangeable, these two TPMS topologies behave statistically equivalently in through-thickness conduction

Mean ΔT difference, with 95% confidence intervals. A CI that crosses zero, grey = fail to reject H_0 .

Red band = $\pm 1.16^\circ\text{C}$ instrument uncertainty.

Green = significant at $\alpha=0.05$

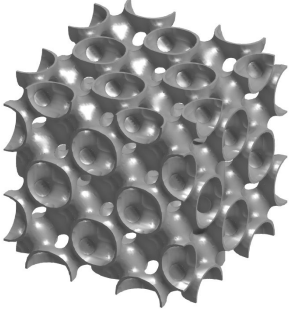
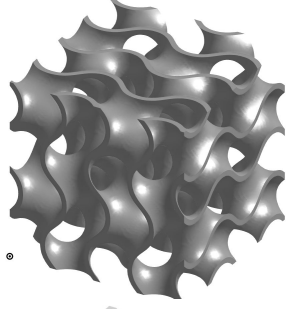
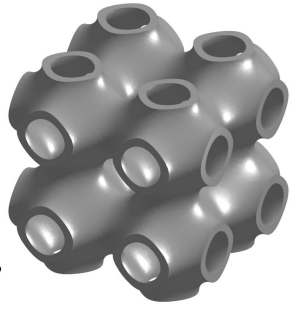
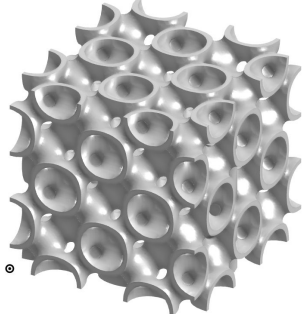
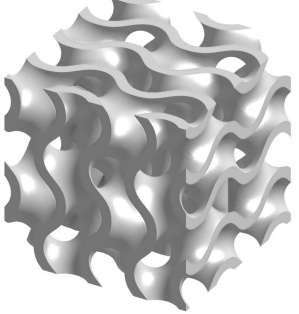
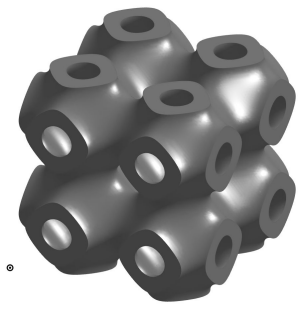
DOE 2 | Conduction Study

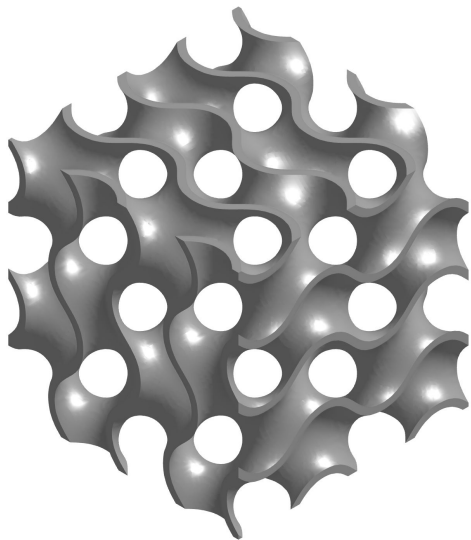
STATISTICAL CONCLUSIONS | HYPOTHESIS CONCLUSION

Verdict: Fail to reject H_0 at $\alpha=0.05$ for all back-face ($\frac{1}{4}$ depth) comparisons, however this topology may not be separable with $n=3$ trials.

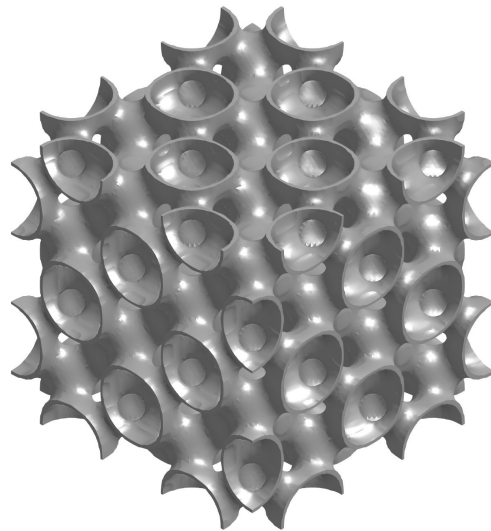
What the data does show:

- One formal rejection: Sleeve vs Solid at $\frac{1}{2}$ depth ($p=0.025$).
- Borderline separation: Sleeve sits 5–6°C cooler than FRD and Gyroid, but fall just above $\alpha=0.05$ at $n=3$
- FRD and Gyroid are equivalent: Indistinguishable at every depth, topology alone does not differentiate the in conduction
- Solid runs hottest, Sleeve coolest at every depth, geometry and fill fraction dominate over lattice topology
- Limitation: At $n=3$, $v \approx 2-4$ and $t_{crit} \approx 2.8-4.1$, best to rerun $n \geq 5$ to resolve.

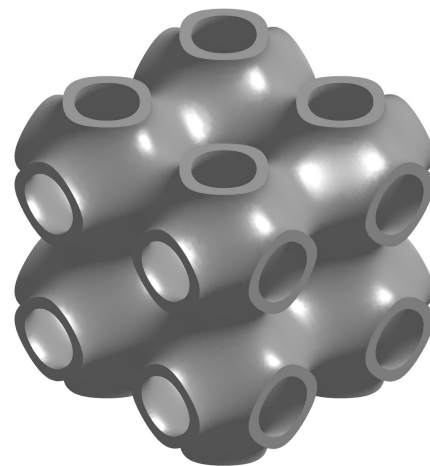
	FRD	Gyroid	Primitive Schwarz
Relative Density = 15			
Relative Density = 30			



Gyroid



FRD



Primitive Schwartz

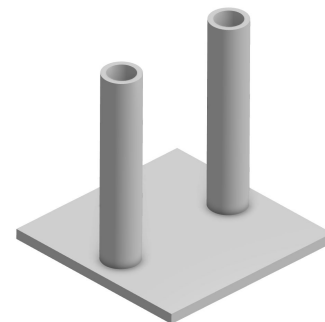
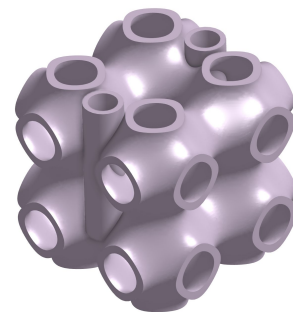
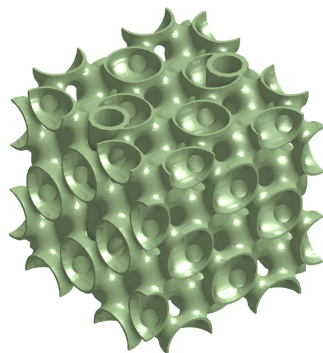
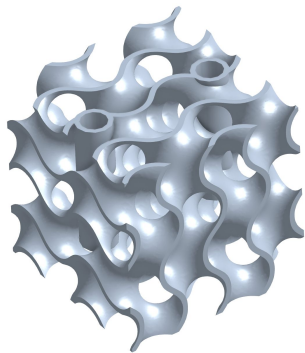
Gyroid

FRD

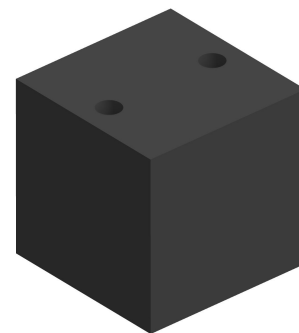
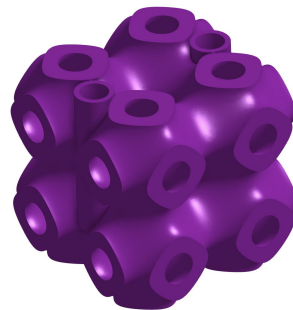
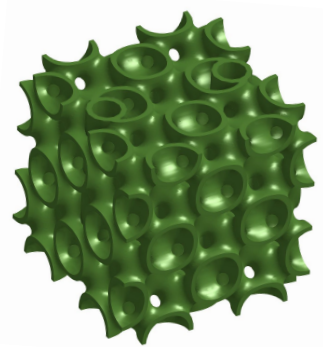
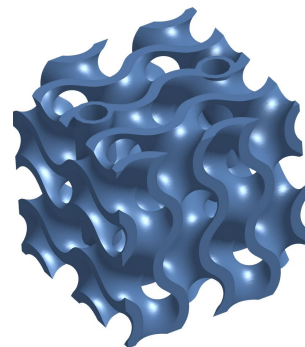
PS

Controls

$\rho_r = 0.15$

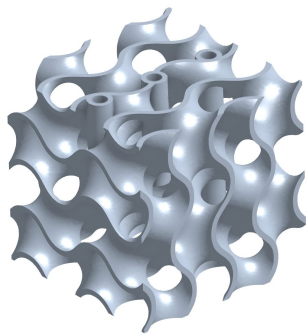


$\rho_r = 0.30$

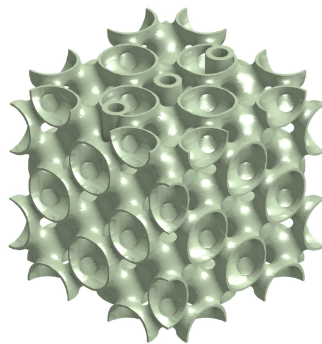


$$\rho_r = 0.15$$

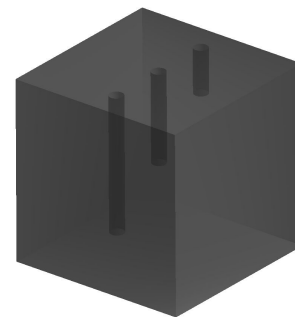
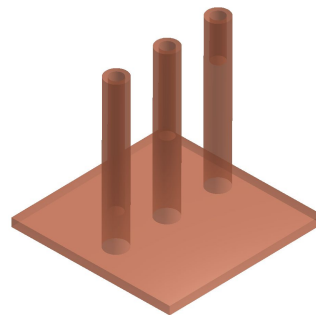
Gyroid



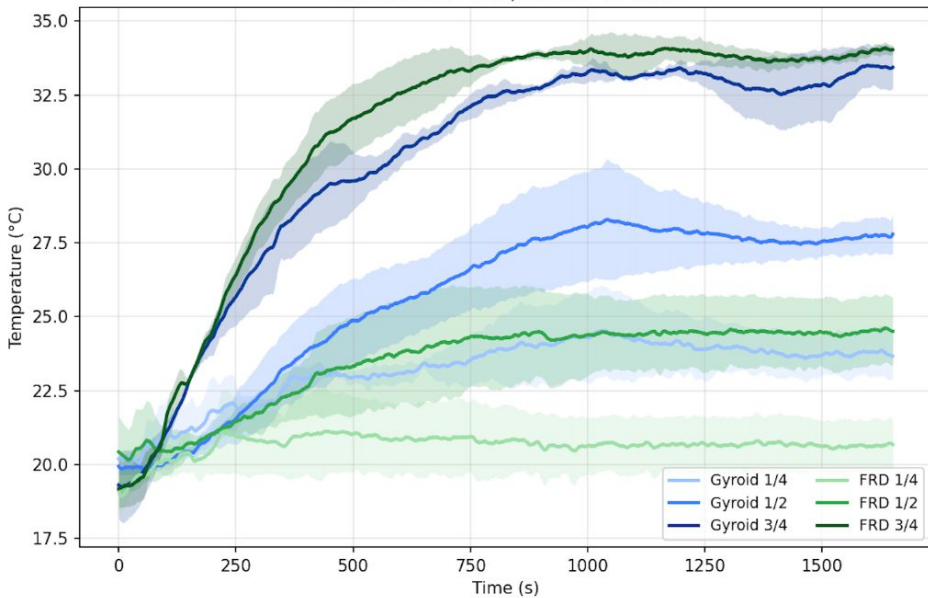
FRD



Controls



GYR and FRD Temperature vs Time



Sleeve and Solid Temperature vs Time

